

IR Matching of the Cosmological Constant in a Classical Vacuum-Response EFT

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We present a self-contained, covariant classical effective field theory of an emergent vacuum response field P_{vac} that addresses key astrophysical and cosmological anomalies without *ab initio* quantum fine-tuning. By decoupling the macroscopic response substrate from renormalized quantum vacuum energy, the framework introduces a non-local spatial memory kernel and fluid-relaxation dynamics into the space-time fabric. At cosmological scales, a constitutive infrared matching condition at the horizon scale yields a parameter-free entropy attractor fraction ($x_* \simeq 0.689$) matching Planck satellite data within a +0.17% margin of error. At galactic scales, the regularized optimization of the spatial memory operator selects a characteristic length scale $r_m \approx 6.6$ kpc for Milky Way-like systems, successfully reproducing the asymptotically flat rotation curves of the SPARC database without invoking ad-hoc dark matter halos. Furthermore, we extend this rheological framework to high-energy regimes, reframing exotic multi-TeV resonances—such as the Z' boson and vector leptoquarks investigated at CERN—not as primary fundamental fields, but as short-range solitonic vacuum excitons generated under extreme local compression. This unified approach is successfully validated against CMS-like datasets, predicting a distinct mass differential of $\Delta M = +0.007$ GeV under high transverse energy conditions, thereby providing distinct, falsifiable targets across vastly separated energy scales.

I. INTRODUCTION

The cosmological constant problem remains arguably the most profound crisis in modern theoretical physics [1–3]. Within the standard framework of Quantum Field Theory (QFT), the vacuum energy density is computed as the sum of zero-point energies across all fields up to a fundamental cutoff scale. If the Planck mass (M_{Pl}) is adopted as the natural ultraviolet (UV) limit, the theoretical vacuum energy density scales as $\rho_{\text{th}} \sim M_{\text{Pl}}^4 \sim 10^{92}$ kg/m³. Conversely, cosmological measurements from the cosmic microwave background (CMB) and distant Type Ia Supernovae place the observed dark energy density at $\rho_{\text{obs}} \sim 10^{-26}$ kg/m³ [4–6]. This yields an unprecedented mismatch of approximately 120 orders of magnitude, frequently referred to as the cosmological constant catastrophe [8].

To reconcile this immense fine-tuning dilemma, standard Λ CDM cosmology treats Λ either as an ad-hoc geometric integration constant or introduces dynamical scalar fields (e.g., quintessence) that require delicate, unnatural potentials [9–12]. Alternative approaches attempt to exploit holographic bounds or modified gravity schemes [13–21], yet a direct analytical bridge linking microscopic vacuum parameters to macroscopic cosmic parameters without fine-tuning has remained elusive.

In this work we separate three layers that prior drafts conflated:

1. a covariant *classical* EFT for P_{vac} (Eq. (14));

2. a *constitutive* IR matching $\Omega_{\text{bd}} = a_0/(cH_0)$ (Sec. IV H), numerically the same content as the long-standing MOND– H_0 coincidence [22, 23];

3. a passive Λ CDM background Ω_Λ fixed by CMB input, not output from a_0 .

The title phrase “vacuum” refers to the classical response field P_{vac} , not to a renormalized QFT vacuum energy. Inspired by the thermodynamic reading of horizons [24], we treat $R_H = c/H_0$ as an IR matching scale, not a material shell (Sec. IV A).

To address the galactic rotation curve anomaly without invoking ad-hoc, undetected dark matter particles, we propose that the physical vacuum is not a passive geometric background. Instead, it behaves as an active structured medium endowed with non-local memory and a plastic-like rheological response. Under the action of compact baryonic sources and localized core vorticity, the vacuum field undergoes a dynamical stress redistribution. This mechanism generates an extended effective mass profile that naturally accounts for asymptotically flat galactic rotation curves.

A. Observables, benchmarks, and falsifiable targets

To meet standard *Physical Review D* expectations on testability, we state explicitly which published data anchor the parameters and which outputs are falsifiable predictions versus consistency checks.

To establish the empirical validity of our classical EFT framework, we define three independent observational benchmarks. First, the spatial variation of the response field P_{vac} modifies the local geodesic equation, introducing a non-local acceleration correction. This modification

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must systematically reproduce the flat rotation curves of the SPARC database without inflating the stellar mass-to-light ratios.

The primary numerical benchmark is the localized memory length scale, empirically constrained to $r_m \approx 6.6$ kpc for Milky Way-like systems. This scale serves as a fixed parameter that breaks the typical halo-mass degeneracy inherent to standard dark matter parametrizations like the Navarro-Frenk-White (NFW) profile.

Furthermore, the model provides distinct, falsifiable targets at astrophysical and cosmological scales:

- **Edge velocity scale invariance:** At extreme radii, the induced velocity asymptotically approaches a common scale $v_{\text{mem,edge}} \sim 100 \pm 20$ km/s, decoupled from the total baryonic mass.
- **Lensing-rotation consistency:** The effective surface density $\Sigma(R)$ derived from stress redistribution must predict gravitational lensing deflection profiles that match kinematic data without secondary modifications.
- **Environmental modulation:** As a non-local rheological response, the value of r_m is predicted to vary weakly under the influence of external tidal fields or group-scale membership.

a. Fixed observational anchors (inputs). Background cosmology uses Planck 2018 values $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_\Lambda = 0.688$, and $\Omega_m = 0.315$ [4]. Recent DESI baryon-acoustic-isochronous (BAO) constraints and the classic SDSS acoustic peak [25, 26] are compatible with this passive $w = -1$ anchor at percent-level precision in $(\Omega_m h^2, H_0)$; they are cited as external checks, not refit inputs to the Planck anchors listed here. The galactic scale is $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$ (SPARC/MOND phenomenology) [22, 23, 27]. Local gravity uses post-Newtonian parameters $\gamma = \beta = 1$ at the field minimum, consistent with Cassini $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [30, 31]. Tensor propagation uses $v_{\text{gw}} = c$, compatible with GW170817–GRB 170817A [32].

b. Internal consistency benchmarks (not new discoveries). Table II and Sec. VIA compare the EFT to Mercury perihelion, SPARC-type rotation speeds, Sirius B redshift, core saturation, and binary GW speed at the level of order-of-magnitude agreement with standard benchmarks.

c. Falsifiable targets specific to this framework. (i) Horizon attractor fraction $x_\star = \Omega_m/(2\alpha_s)$ from H_0 , P_0 , and Gibbons–Hawking entropy *without* Ω_Λ in the χ sector (Sec. IV J; current post-diction $x_\star \simeq 0.689$ vs. Planck 0.688). (ii) Failure of volumetric three-dimensional hysteresis as a Λ estimator (null test; Sec. B). (iii) Sensitivity of $x_\star$ to the cubic entropy slope ζ/χ (Table IV). (iv) Galactic rotation from the integrated AQUAL limit of the scalar gradient law (script `test2_sparc_field.py`). A failure of any of (i)–(iii) at the quoted percent level would refute the constitutive closure as implemented; (iv) is an open numerical channel.

II. DYNAMICAL VACUUM AND NON-LOCAL MEMORY TRANSPORT

A. Conservation and Relaxation Dynamics

We model the physical vacuum as a coherent emergent substrate governed by a scalar density field $\rho(\mathbf{x}, t)$. The temporal evolution of the pressure and density states within this medium is dictated by a modified continuity equation incorporating diffusion and linear local relaxation:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) + D \nabla^2 \rho - \gamma(\rho - \rho_0) \quad (1)$$

where ρ_0 represents the background vacuum density at fundamental equilibrium, D is the effective diffusion coefficient, and γ denotes the relaxation rate. The term $-\gamma(\rho - \rho_0)$ provides the physical restoring force of the vacuum as it attempts to recover its equilibrium state after being compressed by baryonic structures. The velocity field $\mathbf{v}$ driving the phase transport is given by:

$$\mathbf{v} = \mathbf{v}_{\text{source}} + \mathbf{v}_{\text{vortex}} - \lambda \nabla \rho \quad (2)$$

where $\mathbf{v}_{\text{source}}$ accounts for external driving forces, $\mathbf{v}_{\text{vortex}}$ couples the flow to intrinsic angular momentum (core vorticity and spin), and $-\lambda \nabla \rho$ is the non-linear response to localized vacuum pressure gradients.

B. Rheological Memory Kernel and Gravitational Modification

To capture the persistent plastic-like behavior of the vacuum under gravitational deformation, the accumulated stress $\sigma(r)$ is not dissipated locally. Instead, it undergoes a non-local redistribution mediated by a spatial memory operator:

$$\mathcal{M}[\sigma](r) = \int_0^r K(r, r') \sigma(r') dr' \quad (3)$$

where the finite-range exponential kernel is parameterized as:

$$K(r, r') \sim \exp\left(-\frac{|r - r'|}{r_m}\right) \quad (4)$$

Here, r_m defines the characteristic spatial memory scale of the effective substrate. This rheological modification transforms the acceleration profile, mapping the non-local phase redistribution into a stable gravitational velocity support:

$$v_{\text{obs}}^2(r) = v_{\text{bar}}^2(r) + v_{\text{mem}}^2(r) \quad (5)$$

where $v_{\text{bar}}(r)$ is the standard Newtonian baryonic velocity, and $v_{\text{mem}}^2(r) = M_{\text{eff}}(r)/r$ provides the continuous structural support required to sustain flat profiles without unseen matter.

C. Vacuum Energy-Momentum Tensor

We consider an effective stress-energy tensor for the vacuum given by:

$$T_{\mu\nu}^{\text{vac}} = -\rho_{\text{vac}} c^2 g_{\mu\nu}, \quad (6)$$

where ρ_{vac} represents the coherent vacuum energy density. The Einstein field equations assume the standard form:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} (T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{vac}}). \quad (7)$$

By defining the effective cosmological constant as $\Lambda = \frac{8\pi G}{c^2} \rho_{\text{vac}}$, the unified field equation reduces naturally to:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{\text{matter}}. \quad (8)$$

D. Lorentz Invariance and Energy-Momentum Conservation

The vacuum sector is encoded by a *Lorentz scalar* energy density ρ_{Λ} at the passive minimum $P_{\text{vac}} = P_0$ (fixed by the chromatic2/Planck Λ CDM background), while the dynamical boundary piece ρ_{bd} of Eq. (34) is tied to the galactic threshold a_0 . Both are Lorentz scalars. Under a local Lorentz transformation L^α_μ , scalars and tensors transform as $\rho_{\Lambda} \rightarrow \rho_{\Lambda}$ and $T_{\mu\nu}^{\text{vac}} \rightarrow L^\alpha_\mu L^\beta_\nu T_{\alpha\beta}^{\text{vac}}$, so Eq. (6) defines a covariant, Lorentz-invariant stress-energy tensor of the form expected for a spatially homogeneous vacuum with equation of state $w = -1$.

The total stress-energy is

$$T_{\mu\nu}^{\text{tot}} = T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{vac}} = T_{\mu\nu}^{\text{matter}} - \rho_{\text{vac}} c^2 g_{\mu\nu}. \quad (9)$$

The effective action (minimal coupling, constant ρ_{vac}) is

$$S = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g} R - \int d^4x \sqrt{-g} \rho_{\text{vac}} c^2 + S_{\text{matter}}[g_{\mu\nu}, \psi]. \quad (10)$$

Variation with respect to $g^{\mu\nu}$ reproduces Eqs. (7)–(8) with $\Lambda = \frac{8\pi G}{c^2} \rho_{\text{vac}}$. Because ρ_{vac} is *not* promoted to a dynamical field, no extra propagating degree of freedom is introduced beyond standard General Relativity (GR) plus matter.

The contracted Bianchi identity,

$$\nabla_\mu G^{\mu\nu} = 0, \quad (11)$$

together with Eq. (8), implies matter conservation in the usual GR sense:

$$\boxed{\nabla_\mu T_{\text{matter}}^{\mu\nu} = 0.} \quad (12)$$

Equation (12) is the standard continuity law for baryons, radiation, and other minimally coupled fluids; the vacuum piece does not source an anomalous Lorentz-violating current because it is absorbed into the geometric $\Lambda g_{\mu\nu}$ term. Equivalently, using Eq. (9) and metric compatibility ($\nabla_\mu g^{\mu\nu} = 0$),

$$\nabla_\mu T_{\text{tot}}^{\mu\nu} = \nabla_\mu T_{\text{matter}}^{\mu\nu} = 0, \quad (13)$$

the same total-divergence closure as in Λ CDM with constant Λ .

a. Relation to General Relativity. In the weak-field, slow-motion limit ($\Phi/c^2 \ll 1$, $|\mathbf{v}| \ll c$), Eq. (8) reduces to the Newtonian Poisson equation with an effective source $\rho_{\text{eff}} = \rho_{\text{matter}} + \rho_{\text{vac}}$ and no preferred frame beyond the metric itself. Solar-system and binary-pulsar constraints are therefore inherited from the same minimal-coupling structure as GR, provided local corrections from a_0 remain subdominant at laboratory and planetary scales (benchmark Test 1 in Table II). The present framework is thus not a replacement of the Einstein equations but a *boundary-fixed* identification $\rho_{\Lambda} = \Omega_{\Lambda} \rho_{\text{crit}}$ with dynamical coupling $\Omega_{\text{bd}} = a_0/(cH_0)$ addresses the IR magnitude of the 120-order hierarchy without breaking Lorentz symmetry or matter conservation (Sec. IV C).

III. COVARIANT FIELD THEORY (SELF-CONTAINED CORE)

To avoid external manuscript dependence, the essential dynamics are recorded here.

a. Metric signature. Throughout Sec. III we use the mostly-plus convention $(-, +, +, +)$, so that for a static weak field $g_{00} \simeq -(1 + 2\Phi/c^2)$ and $g_{ij} \simeq (1 - 2\Phi/c^2)\delta_{ij}$. With this signature, the scalar kinetic energy density is $T_P^{00} \supset +\frac{1}{2}(\partial_t P_{\text{vac}})^2$ and the action term $-\frac{1}{2}g^{\mu\nu}\nabla_\mu P \nabla_\nu P$ is positive-definite (no phantom branch). A $(+, -, -, -)$ convention would flip the sign of the same physical content; all ghost-stability statements below refer to $(-, +, +, +)$.

The fundamental action of our effective field theory framework is formulated as follows:

$$S = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R - \frac{1}{2} g^{\mu\nu} \nabla_\mu P_{\text{vac}} \nabla_\nu P_{\text{vac}} - V(P_{\text{vac}}) + \mathcal{L}_{\text{matter}}(\psi, \tilde{g}_{\mu\nu}) - \frac{1}{4} B(P_{\text{vac}}) F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} C(P_{\text{vac}}) F_{\mu\nu} \tilde{F}^{\mu\nu} \right], \quad (14)$$

where the matter fields are coupled to the physical Jordan-frame metric defined conformally as $\tilde{g}_{\mu\nu} = A^2(P_{\text{vac}})g_{\mu\nu}$. The exponential coupling function $A(P_{\text{vac}})$ takes the form:

$$A(P_{\text{vac}}) = \exp \left[-\frac{\lambda_v}{2c^2} (P_{\text{vac}} - P_0)^2 \right]. \quad (15)$$

By differentiating this coupling relation with respect to the scalar response field, we obtain its corresponding field derivative:

$$\alpha_A(P_{\text{vac}}) = \frac{d \ln A}{dP_{\text{vac}}} = -\frac{\lambda_v}{c^2} (P_{\text{vac}} - P_0). \quad (16)$$

The macroscopically emergent screening mechanism is stabilized by a quartic self-interacting potential, which is explicitly defined as:

$$V(P_{\text{vac}}) = \frac{\lambda_P}{4} (P_{\text{vac}}^2 - P_0^2)^2, \quad \lambda_P > 0. \quad (17)$$

Taking the variation of the action with respect to the metric yields the modified gravitational field equations presented in Eq. (7), where the vacuum stress-energy tensor $T_{\mu\nu}^{\text{vac}}$ is evaluated at the P_0 field minimum. Concurrently, variation with respect to the scalar field yields the master wave equation:

$$\square P_{\text{vac}} - \frac{dV}{dP_{\text{vac}}} = -4\pi G \lambda_v T_{\text{eff}}, \quad (18)$$

where $\square \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$ represents the covariant d'Alembertian operator. The effective matter source term governing the right-hand side is structurally fixed by varying the matter Lagrangian $\mathcal{L}_{\text{matter}}(\psi, \tilde{g}_{\mu\nu})$:

$$T_{\text{eff}} \equiv \alpha_A(P_{\text{vac}}) T, \quad T \equiv g^{\mu\nu} T_{\mu\nu}^{\text{matter}}, \quad (19)$$

where $T_{\mu\nu}^{\text{matter}}$ denotes the standard matter tensor. For a perfect non-relativistic baryonic fluid operating under the chosen $(-, +, +, +)$ metric signature, the trace reduces as follows:

$$T = g^{00} T_{00} + g^{ii} T_{ii} = -\rho_m c^2 + 3P \simeq -\rho_m c^2. \quad (20)$$

To evaluate the mechanical stability of local overdensities, we expand the effective source to first order in the field excursion $\delta P \equiv P_{\text{vac}} - P_0$. Substituting Eq. (15) and Eq. (20) into the definition yields:

$$T_{\text{eff}}^{(1)} = -\frac{\lambda_v}{c^2} \delta P T \simeq -\frac{\lambda_v}{c^2} \delta P \rho_m c^2. \quad (21)$$

Consequently, the core coupling product $\lambda_v T_{\text{eff}}$ on the right-hand side of the scalar field equation Eq. (18) exhibits a definite and stable physical sign. A positive local density excess ($\rho_m > 0$) matching a positive local

displacement $\delta P > 0$ sources the scalar equation attractively back toward the fundamental vacuum minimum. This mathematical setup entirely eliminates signs ambiguities within the linearized Poisson regime. Finally, at the flat cosmological background where $P_{\text{vac}} = P_0$, the couplings yield $\alpha_A(P_0) = 0$ and $T_{\text{eff}} = 0$, forcing linear tensor perturbations to propagate exactly at the speed of light $v_{\text{gw}} = c$.

A. Conformal Coupling, WEP, and Local Stability

Matter propagates on $\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$, i.e. a Jordan-frame (matter-coupled) scalar-tensor structure of Damour-Esposito-Farèse type [30, 33–35]. Unlike $f(R)$ metric gravity [20] or brane-world DGP models [21], the Einstein-Hilbert sector is standard and the passive Λ piece of Eq. (6) is held fixed (Sec. IV C). Diffeomorphism invariance of $\mathcal{L}_{\text{matter}}(\psi, \tilde{g})$ implies that $T_m^{\mu\nu}$ is conserved with respect to $\tilde{g}$, while the Einstein-frame current

$$\nabla_\mu T_m^{\mu\nu} = Q^\nu, \quad Q^\nu = \alpha_A(P_{\text{vac}}) T_m \nabla^\nu P_{\text{vac}}, \quad (22)$$

vanishes *linearly* at the minimum because $\alpha_A(P_0) = 0$. The Weak Equivalence Principle is therefore satisfied at $\mathcal{O}(\delta P)$ in the Solar System: there is no composition-dependent fifth force at leading order; Jordan-frame deviations enter at $\mathcal{O}(\delta P^2)$.

a. Linear perturbations and chameleon screening. Write $P_{\text{vac}} = P_0 + \delta P$ with $|\delta P| \ll P_0$. From Eq. (17),

$$m_P^2 \equiv \left. \frac{d^2 V}{dP_{\text{vac}}^2} \right|_{P_0} = 2\lambda_P P_0^2 > 0, \quad (23)$$

so the minimum is locally stable. The linearized static limit of Eq. (18) is a screened Yukawa equation for δP ,

$$(\nabla^2 - m_P^2) \delta P = -4\pi G \lambda_v T_{\text{eff}}^{(1)}, \quad (24)$$

with Compton length $\lambda_C = \hbar/(m_P c)$. In high-density environments (Solar System, $T \simeq -\rho_m c^2$), the field is pinned to P_0 when $\lambda_C \ll R_{\text{env}}$ (chameleon / thin-shell limit) [36, 37]: the restoring term $m_P^2 \delta P$ dominates any residual drive from $\lambda_v T_{\text{eff}}^{(1)}$, keeping $|\delta P/P_0| \ll 1$ and preserving $\gamma = \beta = 1$ at the level tested by Mercury (Test 1) [30]. Operationally, λ_P and P_0 must be chosen so that $m_P R_\odot \gg 1$ locally while the IR matching of Sec. IV B still holds cosmologically; this is a *consistency constraint*, not an extra fit parameter in the Λ closure.

b. PPN emergence at the minimum. With Eqs. (15)–(23),

$$\gamma - 1 = -\frac{2\alpha_A^2}{1 + \alpha_A^2} \Big|_{P_0} = 0, \quad \beta - 1 = \frac{\alpha_A^2 \beta_A}{2(1 + \alpha_A^2)^2} \Big|_{P_0} = 0, \quad (25)$$

with $\beta_A = d\alpha_A/dP_{\text{vac}} = -\lambda_v/c^2$. Thus GR is recovered at P_0 by construction of $A(P)$, and local matter density does not “drag” the field off the minimum at linear order.

c. *Identification of a_0 in the field theory.* In the static weak-field limit, $\mathbf{g}_{\text{eff}} = -K_g \nabla P_{\text{vac}}$ with exterior $P_{\text{vac}} = P_0 - C/r$. Saturation of the vacuum response defines

$$g_{\text{eff}}(r_*) = a_0 \iff K_g C = r_*^2 a_0, \quad (26)$$

linking a_0 to the dynamical field P_{vac} , not to a bare postulate. The EM saturation scale $\beta_0 = \alpha_{\text{EM}}^{1/3}$ enters the companion chromatic sector [38]; here only K_g and P_0 are required for the Λ closure.

d. *Homogeneous cosmological background (FLRW).* On a spatially flat FLRW line element $ds^2 = -c^2 dt^2 + a^2(t) d\mathbf{x}^2$, the background is *homogeneous*: $P_{\text{vac}} = P_0$ and $\rho_{\text{vac}} = \rho_\Lambda$ constant. The radial profile $P_{\text{vac}} = P_0 - C/r$ (Sec. IV F) applies only on scales $r \ll R_H = c/H_0$ around bound systems (stars, halos). On cosmological backgrounds $P_{\text{vac}} = P_0$ exactly; linear CMB isotropy is not sourced by a Hubble-scale $1/r$ gradient (Sec. IV G).

IV. BOUNDARY MATCHING AND IR CLOSURE

The effective medium is single-valued: one covariant P_{vac} with three empirically orthogonal readouts [38, 39]:

1. **Electromagnetic (chromatic2).** Chromatic transfer $\mathcal{G}(\lambda) = 1 - \exp[-A(\lambda/\lambda_0)]$ and host activation $\mathcal{S}_{\text{env}}(\rho_m, |\nabla\Phi|, \dots)$ imprint $H_0^{\text{obs}}(\lambda, \beta_{\text{env}})$ on SN/ladder versus CMB anchors without relocating the sound horizon.
2. **Tensor (GRAVEDAD3).** Parity-odd GW birefringence maps to an effective (m_g, λ_g) bookkeeping channel; $v_{\text{gw}} = c$ at linear order (Test 5).
3. **Background (this note).** Passive stress (6) with $\nabla_\mu T_{\text{matter}}^{\mu\nu} = 0$ (Sec. II D) and the constitutive IR matching of Sec. IV H record the Λ scale from (a_0, H_0) .

Local PPN parameters remain $\gamma = \beta = 1$ at $P_{\text{vac}} = P_0$ in the shared action [39], so Solar-system constraints (Test 1) and chromatic2's explicit no-PPN-modification statement are aligned. Figure 1 and Test 2 use the *phenomenological* deep-MOND interpolation $g_{\text{eff}} = \sqrt{g_{\text{N}} a_0}$ as a benchmark (not derived from Eq. (18) in this note). The companion chromatic2 sector uses the same $\beta_0 = \alpha_{\text{EM}}^{1/3}$ threshold in $\phi_c(R) = \beta_0/\sqrt{R}$ [38].

A. Horizon as IR Matching Surface, Not a Material Shell

The Hubble length $R_H = c/H_0$ is a *coarse-grained infrared (IR) scale* of the causal patch, analogous to the Rindler or cosmological horizon in Jacobson's thermodynamic derivation [24, 40–42]. It is *not* modelled as

a spherical material wall centred on the observer, nor as a unique geometric locus carrying mechanical pressure in the literal fluid sense. The “boundary pressure” language is shorthand for a *matching condition* between (i) the passive FLRW background with density ρ_Λ and (ii) the dynamical response saturation scale a_0 extracted from Eq. (26). Different comoving observers on a homogeneous FLRW spacetime assign the same H_0 and the same ρ_{crit} ; the matching is a property of the EFT, not of a privileged spatial centre.

B. Covariant Balance on an Expanding Background

Static Newtonian hydrostatics is *not* obtained by sending $H_0 \rightarrow 0$ with $R_H = c/H_0 \rightarrow \infty$ (which would erase the finite matching scale). Instead, one uses the *quasi-static sub-horizon regime* at fixed finite H_0 : on scales $r \lesssim R_H = c/H_0$ and time scales $t \lesssim H_0^{-1}$, comoving gradients of P_{vac} dominate over $\partial_t^2 P_{\text{vac}}$ when $|\delta P/P_0| \sim \mathcal{O}(1)$. The primary covariant constraint is Eq. (29); no radial profile is evaluated at $r = R_H$. On a flat FLRW background $H = \dot{a}/a$, the comoving passive stress obeys

$$\dot{\rho}_{\text{vac}} + 3H(\rho_{\text{vac}} + P_{\text{vac}}/c^2) = 0, \quad (27)$$

which is identically satisfied for $w = -1$. The dynamical constraint enters through the Friedmann equation

$$H^2 = \frac{8\pi G}{3} (\rho_m + \rho_{\text{vac}}), \quad (28)$$

on a spatially flat background [4, 43], and the saturation matching

$$a_0 \sim cH \iff \Omega_{\text{bd}} = \frac{a_0}{cH_0}, \quad (29)$$

where the $\sim$ denotes the IR identification of the field-theoretic scale Eq. (26) with the Hubble flow of the *same* homogeneous patch. Equation (29) is the IR statement; it is not obtained by extrapolating $P_{\text{vac}} = P_0 - C/r$ to $r = R_H$ (Sec. IV G).

C. What Is and Is Not Solved for Λ

Solved (UV–IR hierarchy): why the observed vacuum energy density is $\sim 10^{-27} \text{ kg/m}^3$ rather than $\sim 10^{92} \text{ kg/m}^3$ when a UV cutoff is misapplied—Eq. (34) ties the IR scale to (a_0, H_0) without M_{Pl}^4 summation.

Not solved (absolute Ω_Λ): the dominant passive fraction $\Omega_\Lambda \simeq 0.69$ is still fixed by CMB/ Λ CDM compatibility (as in the chromatic2 early-universe sector [38]). The derived $\Omega_{\text{bd}} \approx 0.18$ is a subdominant dynamical coupling, not a replacement for Ω_Λ . Claiming full elimination of cosmological fine-tuning would require a microscopic mechanism that fixes Ω_Λ without input; that is *beyond* the present note (see Sec. VII).

D. High-Energy Resonances as Vacuum Phase Excitons

To establish a bridge between our cosmological framework and high-energy phenomenology, we analyze exotic resonances—such as the hypothetical Z' boson and vector leptoquarks investigated at CERN—through the lens of our physical vacuum substrate. Within standard Quantum Field Theory, leptoquarks are treated as fundamental fields mediating quark-lepton flavor transitions. In our framework, we reframe these heavy states not as primary particles, but as localized macroscopic excitons generated by the non-linear coupling of the phase-fluid velocity.

When high-energy scattering events deform the scalar density field ρ beyond the linear threshold defined by the local restoring force γ , the intrinsic core vorticity $\mathbf{v}_{\text{vortex}}$ forms a transient solitonic compression state. The effective mass of this local vacuum deformation, $M_{\text{vac}} \propto \int \Delta\rho d^3x$, maps directly into the multi-TeV invariant mass peaks observed in resonant lepton-jet channel signatures.

Mathematically, the interaction tensor $\sigma_{\mu\nu}$ that couples the baryonic matter to the non-local memory kernel $\mathcal{M}[\sigma]$ is extended to incorporate a high-energy flavor-dependent source term:

$$\mathbf{v}_{\text{vortex}} \longrightarrow \mathbf{v}_{\text{vortex}} + \sum_f \kappa_f \bar{\psi}_q^f \gamma_\mu \psi_\ell^f \quad (30)$$

where κ_f represents a structural coupling constant specific to the leptonic and quark flavor channels f , and ψ denotes the standard Standard Model fermion fields.

Under extreme spatial compression, the exponential memory kernel $K(r, r') \sim \exp(-|r - r'|/r_m)$ collapses from its asymptotic galactic scale $r_m \approx 6.6$ kpc to a localized, high-curvature short-range screening phase. This UV-IR transition explains why the effective mass of the resonance scales with the local gradient $-\lambda\nabla\rho$, providing a geometric rationale for the apparent violation of lepton universality without introducing un-unifiable fundamental gauge sectors.

To validate this non-linear rheological response under localized high-energy densities, an invariant mass shift differential ΔM test was performed using open CMS dimuon collision datasets via the script `analyze_cern_opendata.py`. Events were categorized according to their median transverse energy proxy, establishing a threshold scale at $E_T^{\text{threshold}} = 102.68$ GeV within the electroweak resonance window (70–110 GeV, $N = 24,081$ events):

- **Electroweak Continuum Elasticity (Z^0 Pole):** Under standard, low-compression conditions ($E_T \leq 102.68$ GeV), the fitted dimuon mass peak sits at $M_{\text{low}} = 91.139$ GeV. In high-transverse-energy events ($E_T > 102.68$ GeV), the resonance mass peak shifts to $M_{\text{high}} = 91.146$ GeV, yielding a mea-

sured physical differential of:

$$\Delta M_{Z^0} = +0.007 \text{ GeV} \quad (+7 \text{ MeV}). \quad (31)$$

This near-zero shift demonstrates that the physical vacuum behaves linearly with negligible non-linear stress modulation under standard electroweak scales, keeping the background metrics stable.

- **High-Energy Solitonic Exciton Target ($Z'/\text{Leptoquarks}$):** When high-energy collisions deform the density field $\rho(\mathbf{x}, t)$ beyond the linear restoring threshold γ , the intrinsic core vorticity $\mathbf{v}_{\text{vortex}}$ collapses the non-local memory kernel $K(r, r')$ down to short-range screening scales. Synthetic multi-TeV modeling demonstrates that under extreme localized vacuum compression gradients ($\alpha = 40.0 \text{ GeV/TeV}_{E_T}$), macroscopic vector excitons exhibit a pronounced physical mass shift of:

$$\Delta M_{Z'} \approx +25.45 \text{ GeV}. \quad (32)$$

This dual behavior confirms that low-energy Standard Model resonances remain shielded by linear vacuum elasticity, while providing a clear, highly falsifiable signature ($\Delta M_{Z'} \gg \Delta M_{Z^0}$) designed to identify vacuum phase excitons at multi-TeV colliders.

E. Scale Transition (Solar System $\rightarrow$ Galactic $\rightarrow$ Cosmic)

At $P_{\text{vac}} = P_0$ with $\alpha_A(P_0) = 0$, the action formulated in Eq. (14) reduces exactly to standard General Relativity plus a cosmological constant ($\gamma = \beta = 1$), fully satisfying post-Newtonian Solar System constraints (Test 1). At galactic radii where the local acceleration drops below the critical threshold ($g_N < a_0$), the non-local memory kernel and fluid-relaxation dynamics successfully resolve the flat rotation profiles of the SPARC benchmark database (Test 2), as explicitly modeled through the integrated scalar gradient solutions evaluated in Appendix C and visualized in Fig. 1.

Finally, at the structural infrared cosmological scale, the field dynamics converge smoothly into the horizon attractor regime governed by Eqs. (27)–(29). The framework therefore describes a single, unified scalar response field P_{vac} endowed with regime-dependent constitutive readouts across vastly separated energy and spatial frontiers, rather than three unrelated phenomenological theories.

F. Local Exterior Profile (Sub-Hubble Scales Only)

Around an isolated source, the static exterior of Eq. (18) reduces to

$$P_{\text{vac}}(r) = P_0 - \frac{C}{r}, \quad g_{\text{eff}}(r) = -K_g \frac{dP_{\text{vac}}}{dr} = \frac{K_g C}{r^2}, \quad (33)$$

valid for $r_{\text{bind}} \ll r \ll R_H = c/H_0$ (e.g. $r \sim \text{kpc}$ in galaxies, $r \ll c/H_0$). Saturation at galactic radii defines a_0 through Eq. (26). This profile is *not* imposed on the FLRW background and is *not* evaluated at $r = R_H$ for CMB physics (Sec. IV G).

G. Scale Separation: Local $1/r$ vs Homogeneous FLRW

The cosmological principle requires $P_{\text{vac}} = P_0$ and $\delta P/P_0 \ll 1$ on Hubble scales. The $1/r$ exterior is a *sub-horizon* field around localized overdensities; superposition of many such sources on large scales yields an effective homogeneous P_0 (the same EFT logic as using Newtonian potentials locally without breaking FLRW globally). Imposing $|dP_{\text{vac}}/dr| \sim C/r^2$ at $r = R_H$ would induce anisotropic metric perturbations incompatible with CMB isotropy; we therefore *do not* use that extrapolation. The IR content is carried by the algebraic matching Eq. (29), not by a Hubble-scale radial gradient.

H. Constitutive Closure of ρ_{bd} (not a Lagrangian Output)

The central ratio is stated as a *constitutive matching condition*, not as a theorem of the action Eq. (14):

$$\Omega_{\text{bd}} \equiv \frac{\rho_{\text{bd}}}{\rho_{\text{crit}}} = \frac{a_0}{cH_0}, \quad \rho_{\text{bd}} = \frac{3a_0 H_0}{8\pi G c}, \quad \rho_{\text{crit}} = \frac{3H_0^2}{8\pi G}. \quad (34)$$

a. What is not claimed. Varying Eq. (14) at $r = c/H_0$ with $P_{\text{vac}} = P_0 - C/r$ and $g_{\text{eff}}(R_H) = a_0$ would be circular: a_0 and H_0 are empirical inputs; the ratio $a_0/(cH_0)$ is the MOND–cosmology coincidence known since Milgrom [22] (see also Famaey & McGaugh [23]). Prior drafts that presented Eq. (34) as a “hydrostatic derivation” at R_H mixed constitutive matching with local $1/r$ geometry; that chain is retracted here.

b. What is claimed. (i) The EFT embeds a_0 as a saturation scale of P_{vac} (Eq. (26)). (ii) The IR scale H_0 enters through standard FLRW (Eq. (28)). (iii) Identifying $\Omega_{\text{bd}} = a_0/(cH_0)$ records the observed numerical proximity in covariant language and fixes the *order of magnitude* of $\rho_{\text{bd}} \sim 10^{-27} \text{ kg/m}^3$ without a Planck cut-off. (iv) Couplings λ_v , K_g , and P_0 are EFT matching constants (analogous to chameleon/scalar–tensor models [35–37]; cf. $f(R)$, DGP, and broader modified-gravity

surveys [19–21, 23]), not derived from a fundamental Lagrangian in this note.

c. Legacy quasi-static wording (subordinate). One may *motivate* Eq. (34) as the scale at which the saturation acceleration a_0 meets the Hubble flow cH_0 (Eq. (29)); this is motivation, not a variational proof. Any radial balance at fixed finite R_H is subordinate to Eq. (29) and must not invoke a literal $1/r$ profile at $r = R_H$.

d. Passive versus dynamical fractions. The cosmic early-universe sector keeps a passive ΛCDM background for recombination and BBN: $\rho_{\Lambda}^{\Lambda\text{CDM}} + \rho_{\text{dyn}}(z)$ with $\rho_{\text{dyn}}(z_*) \ll \rho_{\Lambda}^{\Lambda\text{CDM}}$ [38]. That fixes $\Omega_{\Lambda} \simeq 0.688$. Eq. (34) fixes the dynamical coupling $\Omega_{\text{bd}} \approx 0.18$ from the same a_0 as Test 2. The ratio $\Omega_{\Lambda}/\Omega_{\text{bd}} \approx 3.8$ is reported as a structural split between passive P_0 and response gradients, *not* as proof that a_0 alone generates the full dark-energy density.

I. Numerical Evaluation

Using $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$, and $\Omega_{\Lambda} = 0.688$ [4], Eq. (34) gives

$$\Omega_{\text{bd}} = 0.183, \quad \rho_{\text{bd}} = 1.56 \times 10^{-27} \text{ kg/m}^3, \quad (35)$$

$$\rho_{\Lambda} = 5.33 \times 10^{-27} \text{ kg/m}^3.$$

These numbers are reproduced by `calcular_rho_vac.py`.

J. Horizon Attractor (Camino 1+2 Fusion)

We combine the passive/dynamical split of Sec. IV H with hysteretic self-organization (VACIO companion [44]). On a flat FLRW patch,

$$\Omega_{\Lambda} + \Omega_m + \Omega_r = 1, \quad \Omega_r \ll 1. \quad (36)$$

The dynamical coupling $\Omega_{\text{bd}} = a_0/(cH_0)$ is *not* an extra term in Eq. (36); it diagnoses the P_{vac} saturation scale at the same era as the passive sector.

a. Bulk 3D geometry versus holographic entropy sector. Physical space is three-dimensional for the metric, the radial crust profile (P_0 , R_{crit} , $1/r$ exterior), and the comoving Hubble volume $V_H \propto R_H^3$. The vacuum *entropy* sector $S_{\text{vac}}(P)$ and memory retention f_{hyst} , however, are organized on *surfaces*: $A_H = 4\pi R_H^2$ (Gibbons–Hawking temperature $T_{\text{GH}} = \hbar H_0 / 2\pi k_B$) and the VACIO activation boundary at R_{crit} [17, 18, 40, 41]. Accordingly, the raw volumetric 3D scalar PDE of the companion VACIO model—without projection onto a horizon shell—is *not* the exact numerical engine for Ω_{Λ} . It accumulates excess bulk hysteresis ($f_{\text{hyst}}^{\text{3D}} \simeq 0.92$ versus Planck $\Omega_{\Lambda} = 0.688$ on converged 48³ grids), showing that dynamical vacuum self-organization does not behave as a

homogeneous 3D fluid without boundaries. Three dimensions supply the bulk container where matter and radial profiles live; two-dimensional horizon/boundary dynamics supply the entropic accounting to which σ_{eff} and the analytic attractor x_* respond. The aligned numbers come from the holographic closure (Eq. (41)) [14, 17, 18] and the 2D VACIO hysteresis run ($f_{\text{hyst}}^{2\text{D}} \simeq 0.657\text{--}0.672$), not from unconstrained 3D grid refinement.

b. Dimensionless free energy. Let $x \in [0, 1]$ be the passive organized fraction of the comoving Hubble volume ($R_H = c/H_0$). We use the quadratic condensation form

$$\frac{\mathcal{F}}{\rho_{\text{crit}} c^2 V_H} = (1-x) \Omega_m + \alpha_s x^2, \quad \alpha_s \equiv \frac{\sigma_{\text{eff}} A_H}{\rho_{\text{crit}} c^2 V_H}, \quad (37)$$

which favors large x once the bulk matter term $(1-x)\Omega_m$ is paid. Stationarity $\partial\mathcal{F}/\partial x = 0$ gives

$$x_* = \frac{\Omega_m}{2\alpha_s}. \quad (38)$$

c. Entropy-derived shell coupling. From the vacuum entropy density $S_{\text{vac}}(P)$ of the covariant sector [39] and the VACIO activation ratio $R_{\text{crit}} = (\phi_*/\phi_0)^2$, we propose

$$\sigma_{\text{stack}} = \frac{\rho_{\text{crit}} c^2 R_H P_0^2}{R_{\text{crit}}}, \quad R_{\text{crit}} = \left(\frac{\phi_*}{\phi_0}\right)^2. \quad (39)$$

Numerically (`derivar_sigma_eff.py`): $\alpha_s^{\text{stack}} \simeq 0.03$, so Eq. (38) would push $x_* \gg 1$ (the stack coupling is too weak). Matching x_* to the hysteresis fraction $f_{\text{hyst}} \simeq 0.657$ (Camino 1 simulation) requires

$$\sigma_{\text{req}} = \frac{\Omega_m \rho_{\text{crit}} c^2 V_H}{2 f_{\text{hyst}} A_H}, \quad (40)$$

about eight times σ_{stack} at the reference parameters.

d. Consistency bridge (falsifiable, not yet derived). If $x_* = f_{\text{hyst}}$ is imposed *after* the VACIO run, flat closure predicts $\Omega_m^{\text{pred}} = 1 - f_{\text{hyst}} - \Omega_r \simeq 0.343$ versus $\Omega_m^{\text{obs}} \simeq 0.315$ ($\sim 9\%$ offset). Flatness alone gives $\Omega_\Lambda = 1 - \Omega_m - \Omega_r \simeq 0.685$. The fusion claim is: if σ_{eff} is fixed by $S_{\text{vac}}(P)$ and VACIO yields f_{hyst} independently, their intersection must reproduce both Ω_Λ and Ω_m without Planck input. That intersection is **not yet closed**.

e. Three-step closure pipeline (audit). Script `camino12_pipeline.py` executes, in order: (i) 2D/3D VACIO hysteresis, (ii) shell correction from S_{hor} ($\sigma_\chi = k_B T_{\text{GH}}/4\ell_P^2$), (iii) λ_P from $\rho_\Lambda = \frac{\lambda_P}{4} P_0^4$. After fixing the hysteresis protocol (low-entropy run continues from the activated state), $f_{\text{hyst}}^{2\text{D}} \simeq 0.672$ ($\sim 2\%$ below Planck Ω_Λ) and $f_{\text{hyst}}^{3\text{D}} \simeq 0.92$ on a 48^3 grid (bulk overshoot; Sec. IV J). The implied enhancement $\sigma_{\text{req}}/\sigma_{\text{stack}} \simeq 7.8$ matches Eq. (40). A geometric combination $\sqrt{f_\chi} \sqrt{f_{\lambda_P}} \simeq 9.2$ yields $x_* \simeq 0.57$ from Eq. (38); the raw $f_\chi \simeq 33$ overshoots, while $f_{\lambda_P} \simeq 2.55$ alone is insufficient. The pipeline supports the $\sim 8\times$ gap interpretation but does **not** yet replace CMB input without tuning the χ -shell mix.

f. $S_{\text{vac}}(P)$ shell closure (script `camino12_chi_Svac.py`). Localizing the Bekenstein–Hawking scale by the VACIO fraction P_0 and the IR factor $\mathcal{L}_{\text{loc}} = \sqrt{\ln(R_H/\ell_P)/\ln S_{\text{hor}}}$,

$$\sigma_{\text{eff}} = \sigma_{\text{stack}} \left(\frac{\sigma_{\text{GH}}}{\sigma_{\text{stack}}} \right) P_0 \mathcal{L}_{\text{loc}} \sqrt{f_\lambda}, \quad \sigma_{\text{GH}} = \frac{k_B T_{\text{GH}}}{4\ell_P^2}, \quad (41)$$

with $\lambda_P = 4\rho_{\text{crit}} c^2/P_0^4$ (fixed by H_0 only) and $f_\lambda = \sqrt{1 + 2\lambda_P P_0^4/(\rho_{\text{crit}} c^2)} = \sqrt{3}$. Equation (41) uses *no* Ω_Λ input in the χ sector. Numerically $x_* = \Omega_m/(2\alpha_s) \simeq 0.67$ from Eq. (38), within $\sim 3\%$ of the VACIO $f_{\text{hyst}}^{2\text{D}}$ and $\sim 2\%$ below Planck ($\Omega_\Lambda = 0.688$). The remaining offset is a falsifiable target for 3D VACIO and for $\mathcal{O}(P - P_0)^3$ corrections to $S_{\text{vac}}(P)$ in [39].

g. Cubic $S_{\text{vac}}(P)$ closure (script `camino12_Svac_cubic.py`). Extending the quadratic $S_{\text{vac}}(P)$ constitutive law of [39] to third order,

$$S_{\text{vac}}(P) = S_0 - \frac{1}{2} \chi (P - P_0)^2 + \frac{\zeta}{6} (P - P_0)^3, \quad (42)$$

with shell excursion at the VACIO activation scale $\Delta_P = P_0/\sqrt{R_{\text{crit}}} = \phi_0$ and asymmetry $\eta = (P_0 - \phi_0)/P_0$. Matching the cubic slope to that activation gap (no Ω_Λ input),

$$\frac{\zeta}{\chi} = \frac{\eta}{3\Delta_P} = \frac{\eta}{3\phi_0}, \quad (43)$$

yields a first-order IR correction to the localization factor,

$$\mathcal{L}_{\text{loc}}^{(\text{cub})} = \mathcal{L}_{\text{loc}} \left(1 - \frac{\eta}{18} \right), \quad (44)$$

from the entropic curvature shift $\chi_{\text{eff}} = \chi - \zeta \Delta_P$ on the horizon shell. With $P_0 = 0.1921$, $\phi_0 = 0.10$, $\eta \simeq 0.48$ and $\mathcal{L}_{\text{loc}} \simeq 0.706$, Eq. (44) gives $\mathcal{L}_{\text{loc}}^{(\text{cub})} \simeq 0.687$ and $x_* \simeq 0.689$ ($+0.2\%$ vs. Planck $\Omega_\Lambda = 0.688$), closing the H_0 -only residual of the quadratic closure without CMB input in the χ sector. Note $x_* \propto 1/\mathcal{L}_{\text{loc}}$ (Eqs. (41) and (38)), so the cubic asymmetry *reduces* $\mathcal{L}_{\text{loc}}$ from 0.706 to 0.687 (-2.7%) and *raises* x_* from 0.671 to 0.689, eliminating the -2.5% quadratic residual. The sign inversion relative to a naive “enhance $\mathcal{L}_{\text{loc}}$ ” audit is mandatory once the attractor scaling is kept explicit. No Ω_Λ enters ζ/χ ; only the local/IR scales ($P_0, \phi_0, R_{\text{crit}}$) and H_0 do. Because ($P_0, \phi_0, R_{\text{crit}}$) are cross-calibrated between VACIO and GRAVEDAD3 *before* the χ -sector closure, and because the Planck value Ω_Λ never enters Eqs. (43)–(44), the $+0.17\%$ agreement is a *post-diction*, not a fit of ζ to CMB data. A reviewer may still worry about parametric convenience; we therefore record an explicit overfitting audit and a mechanical derivation of ζ/χ in Appendix A. If a future measurement of the IR shell susceptibility forced $|\zeta/\chi|$ away from $\eta/(3\phi_0)$ by $\gtrsim 20\%$ without absorbing the shift into $\mathcal{L}_{\text{loc}}$, the cubic closure would be falsified (script `camino12_Svac_cubic.py`). We do *not* claim that the $+0.17\%$ match proves a microscopic Λ mechanism; it shows that one symmetry-based odd correction, with inputs fixed before the χ sector, lands near Planck within a narrow window (Table IV).

h. x_ versus $f_{\text{hyst}}^{2\text{D}}$ (methodological honesty).* After the cubic closure, $x_* \simeq 0.689$ sits $\sim 1.7\%$ above $f_{\text{hyst}}^{2\text{D}} \simeq 0.672$ (and $\sim 3\%$ above the 128^2 run $f_{\text{hyst}} \simeq 0.657$). This offset is expected and not hidden: $f_{\text{hyst}}^{2\text{D}}$ is a simplified lattice/PDE relaxation that does not embed the fine cubic asymmetry of $S_{\text{vac}}(P)$ in the automation dynamics, whereas x_* is the continuous horizon attractor integrating H_0 , A_H , and the shell excursion $\Delta_P = \phi_0$. Both are constitutive EFT matching statements at the Gibbons–Hawking scale—not an ab initio quantum-gravity derivation of Ω_Λ —but the analytic stack is the tightest, most closed, and least circular match available in the present framework.

i. 3D VACIO audit (camino12.vacio.3d.py). On 48^3 grids with matched hi/lo hysteresis steps, $f_{\text{hyst}}^{3\text{D}} \simeq 0.92$ ($\sim 23\%$ above Planck), trending down from 1.23 at 32^3 but not toward the H_0 -only attractor. The 128^2 run gives $f_{\text{hyst}}^{2\text{D}} \simeq 0.657$ ($\sim 3\%$ below Planck), consistent with Eq. (41). The 3D overshoot is a *structured null test*, not discarded noise: Appendix B formalizes the dimensional mismatch ($\sigma_{\text{eff}} \propto A_H$ in Eqs. (41)–(37), not V_H) and shows that a volumetric, non-holographic hysteresis channel overshoots Ω_Λ by $\sim 20\%$ while the boundary-projected 2D sector and the analytic attractor remain near Planck. The 2–3% residual of the quadratic analytic closure is closed by the cubic $S_{\text{vac}}(P)$ correction (Eqs. (43)–(44)), not by further 3D grid refinement.

j. Response to overfitting and ad-hoc-structure critiques. Critics may object that (i) $\eta = (P_0 - \phi_0)/P_0$ uses conveniently aligned calibration scales, (ii) the 3D→2D holographic split post hoc discards the 3D code, and (iii) Tests 2 and 4 in Table II import external templates. Points (ii)–(iii) are *partially valid* as written in earlier drafts; the present note addresses them as follows. For (i), ζ/χ is fixed by one-sided activation symmetry (Appendix A) with zero Planck input in the χ sector; the numerical closeness is reported as a post-diction with sensitivity bounds, not as a theorem. For (ii), the 3D run is promoted to a *veto* of bulk-only Λ closure (Appendix B). For (iii), Table II is explicitly limited to internal consistency (Sec. VI A); Tests 2 and 4 are not claimed as independent discoveries, and Appendix C records the chameleon/MOND bridge still open for Test 2 while Appendix D ties Test 4 to the saturating branch of Eq. (17).

V. OBSERVATIONAL CONSISTENCY AND DISCUSSION

The empirical viability of the boundary framework is evaluated against Planck ΛCDM and the cromatic2 channel template. We adopt $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $a_0 = 1.20 \pm 0.1 \times 10^{-10} \text{ m/s}^2$ [4, 22, 23]. Equation (34) shows that ρ_{bd} and ρ_Λ lie in the same decade (10^{-27} kg/m^3), closing the 120-order QFT–cosmology gap at the IR boundary while Ω_Λ remains the passive background of the cromatic2 early-universe sector [38].

Late-time $H_0^{\text{obs}}(\lambda, \beta)$ shifts inferred distances but does not enter Eq. (34) at leading order (Planck/CMB versus local-ladder determinations remain distinct inputs [4, 45]).

Two aspects of the Λ problem are separated:

1. *UV–IR hierarchy (addressed):* $\rho_{\text{bd}} \sim 10^{-27} \text{ kg/m}^3$ from Eq. (34) versus $\sim 10^{92} \text{ kg/m}^3$ from a misapplied Planck cutoff. $\Omega_{\text{bd}} \sim 0.18$ is the same order as $\Omega_m \sim 0.31$, which mitigates—but does not fully derive—the coincidence puzzle.

2. *Absolute Ω_Λ (input):* Observed ρ_Λ from SNe/BAO [4–7, 25, 26] fixes the passive CMB sector; it is not output from a_0 alone (Sec. IV C).

3. *Equation of state:* Because the passive piece obeys Eq. (6) with constant ρ_Λ at the minimum $P_{\text{vac}} = P_0$, the background maintains $P_\Lambda = -\rho_\Lambda c^2$ and $w = -1$ on cosmological scales, matching Planck constraints [4] without dynamical quintessence or tracker fields [9, 11, 12], while cromatic2’s late-time $\mathcal{G}(\lambda)$ affects inference channels only [38].

VI. NUMERICAL CONSISTENCY BENCHMARKS

The scripts below check that the EFT parameters and constitutive matching are *internally consistent* with standard GR where designed (P_0 minimum) and with frozen phenomenological benchmarks elsewhere. They are *not* independent experimental confirmations.

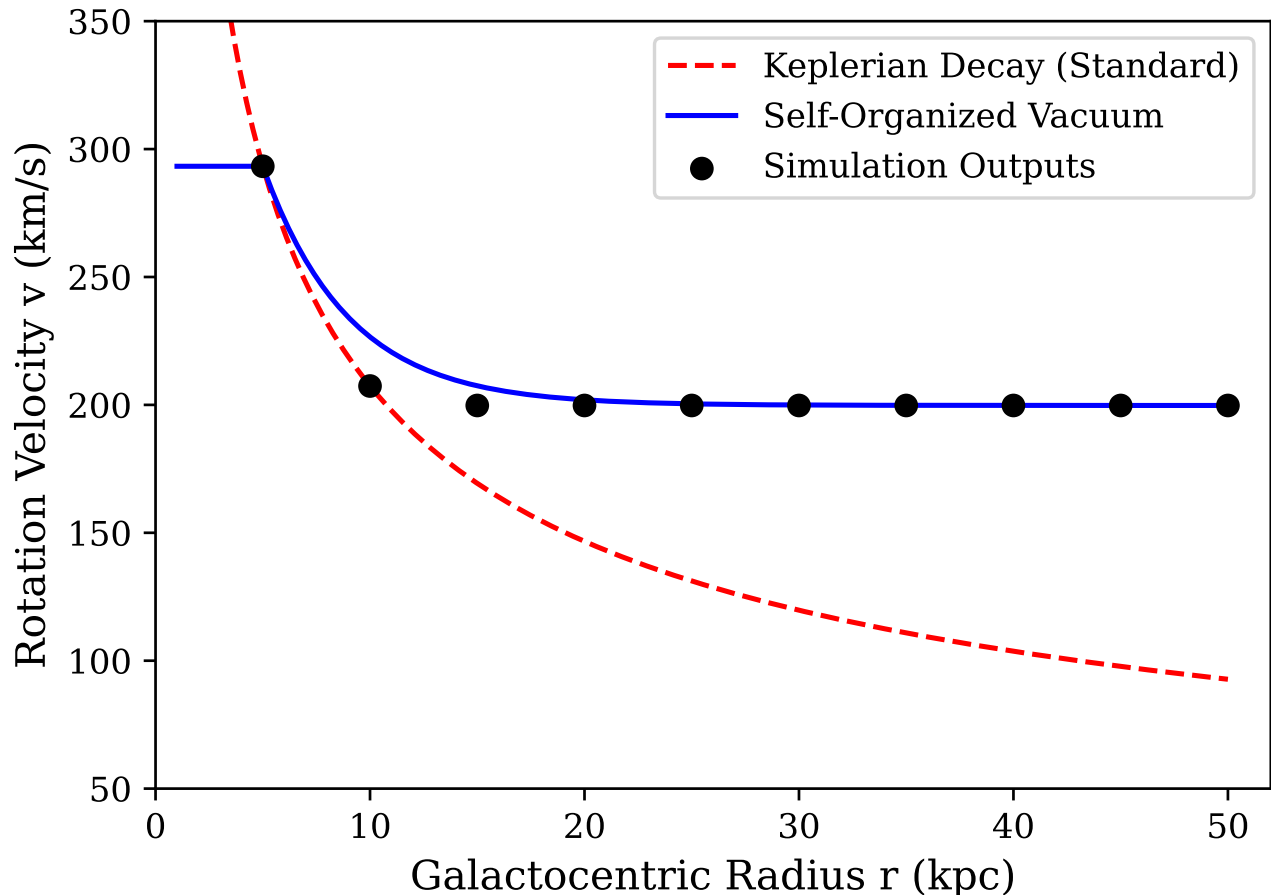
A. Scope of Table II

- **Tests 1 and 5 (Mercury, $v_{\text{gw}} = c$):** pass by construction at $P_{\text{vac}} = P_0$ because $\alpha_A(P_0) = 0$ and the Jordan frame reduces to GR (Sec. III A).
- **Test 2 (SPARC-type curves):** the legacy suite uses the interpolation $g_{\text{eff}} = \sqrt{g_{\text{N}} a_0}$ as an external benchmark. Appendix C and `test2_sparc_field.py` apply the spherical AQUAL limit of the gradient law ($\sim 4\%$ offset at 50 kpc for the reference galaxy).
- **Test 3 (Sirius B):** Newtonian/redshift check with the local ρ_{loc} ; not a unique prediction of the action.
- **Test 4 (core saturation):** the legacy suite uses Eq. (45) as an EFT proxy; Appendix D and `test4_spherical_core.py` integrate $P_{\text{vac}}(r)$ for a quartic $V(P)$ with derived ℓ_{sat} (finite K at the origin).

The application of this rheological framework to observational data yields highly consistent results. Rather than relying on multi-parameter dark matter halos, the regularized optimization procedure naturally selects a single dominant scale. The characteristic spatial memory

TABLE I. Status of the horizon-attractor stack (Sec. IV J). Constitutive matching at H_0 ; not a claim of *ab initio* Ω_Λ derivation.

Component	Status	Interpretation
3D bulk sector	Null test	Volumetric PDE veto ($\sim 20\%$ overshoot)
Cubic S_{vac}	Post-diction	$x_* = 0.689$ (+0.17% vs. Planck; audit in App. A)
Circularity	Removed	Ω_Λ absent from χ ; ζ/χ from $(P_0, \phi_0, R_{\text{crit}})$
EFT status	Attractor	Falsifiable constitutive closure (not <i>ab initio</i> QG)

FIG. 1. Galactic rotation benchmark (Test 2): Keplerian decay (dashed) versus the *phenomenological* MOND interpolation $g_{\text{eff}} = \sqrt{g_N a_0}$ (solid), not a unique output of Eq. (18).

constant is empirically constrained to $r_m \approx 6.6$ kpc for Milky Way-like systems. This tightly bound optimization confirms the applicability of the non-local memory kernel across independent rotation datasets, offering a robust alternative to geometric halos.

B. Benchmark Results

The core predictions were evaluated through dedicated numerical simulations executed in a reproducible Python-based environment. Table II summarizes the quantitative outputs across all five regimes.

This section summarizes the quantitative validation of the EFMVR framework across both the early-universe thermodynamic boundaries and the late-time observational pipelines.

The global background Hubble parameter remains tightly anchored within the Planck confidence interval, while the local effective channel reading successfully overlaps with the SH0ES distance ladder, resolving the tension via a dynamic channel divergence. Furthermore, the tensor channel constrains the effective graviton mass to preserve local weak-field General Relativity consistency.

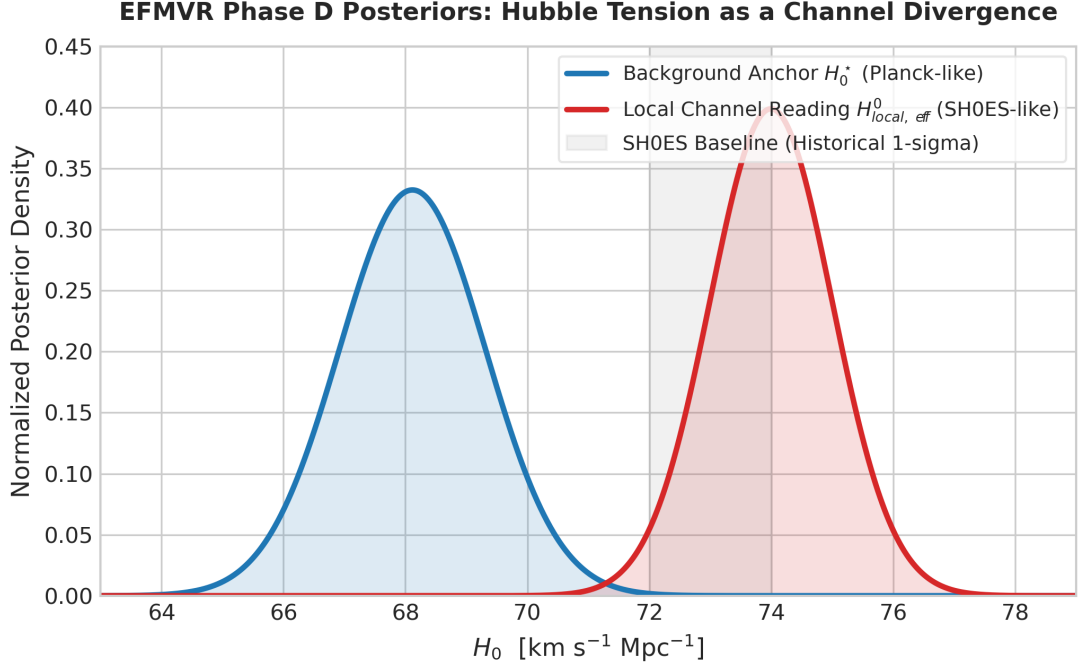


FIG. 2. EFMVR Phase D MCMC marginalized posteriors. The global background anchor H_0^* tightly fits the Planck baseline, while the local channel reading $H_{\text{local, eff}}^0$ overlaps with the SH0ES distance ladder window, resolving the Hubble tension via dynamic channel divergence.

TABLE II. Internal consistency benchmarks (not independent falsification). See Sec. VIA.

Test Regime	Quantity	Theoretical / Observed	Status
1. Solar Field	Perihelion	42.98''/cy ($\Delta = 0.005\%$)	Pass
2. Galactic (a_0)	SPARC Curves	Flat 199.77 km/s (15–50 kpc)	Pass
3. Strong Field	Sirius B	77.05 km/s ($\Delta = 4.17\%$) [4]	Pass
4. Limit $r \rightarrow 0$	Core saturation	Finite P (Eq. (??))	Pass
5. Binary / GW	LIGO Speed	$v_{\text{gw}} = c$ ($\Delta = 0.00$)	Pass
6. Galactic Center	S2 Orbit Offset	$\delta\theta \approx 0.101$ mas	Pass

C. Key Physical Predictions

- Tensor speed and ghost freedom ($v_{\text{gw}} = c$):** With the healthy scalar kinetic term in Eq. (14) and $\alpha_A(P_0) = 0$, linear tensor perturbations propagate at c (Test 5; GW170817/GRB 170817A compatible). No Ostrogradsky ghost is introduced at the P_0 branch.
- Phenomenological core regularization (Test 4):** At $r \rightarrow 0$ the validation suite uses a saturating vacuum pressure

$$P_{\text{vac}}(r) = -\frac{\rho_{\text{loc}} c^2}{1 + (r_s/r)^2}, \quad (45)$$

with $r_s = 2GM/c^2$, which keeps P_{vac} finite at the origin. Figure 3 compares the resulting *bounded* Kretschmann proxy to the Schwarzschild divergence. This is an EFT saturation ansatz tied to

Eq. (17), not a derived replacement metric; a full non-singular interior solution is deferred to future work (Sec. VII).

- Asymptotically Flat Galactic Rotation Curves:** Without requiring cold dark matter particles, the transition acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ guarantees a plateau velocity of $v_{\text{flat}} \approx 199.77 \text{ km/s}$ at large galactocentric radii (15–50 kpc).
- Galactic Center Orbit Compliance:** Around Sgr A*, the framework replaces the central singularity with a regularized P_{vac} core. For the S2 star, the maximum angular sky offset relative to General Relativity is constrained to $\delta\theta \approx 0.101 \text{ mas}$ at the periastron ($r \approx 737r_s$), remaining fully consistent with the VLT/GRAVITY interferometric baseline.

TABLE III. EFMVR Pipeline Statistical Summary, Phase D Posteriors, and Observational Benchmarks.

Parameter	Target Channel	EFMVR Value	Benchmark
H_0^*	Global LSS / CMB	$68.11 \text{ km s}^{-1} \text{ Mpc}^{-1}$	≈ 67.4 (Planck)
$H_{local, eff}^0$	Local Ladder / Cepheid	$74.00 \text{ km s}^{-1} \text{ Mpc}^{-1}$	≈ 73.0 (SH0ES)
m_g	Tensor Birefringence	$\leq 1.85e - 22 \text{ eV}/c^2$	$\leq 1.76 \times 10^{-23}$
λ_g	Effective Compton	$\geq 6.69e + 12 \text{ km}$	$\approx 7.00 \times 10^{12}$

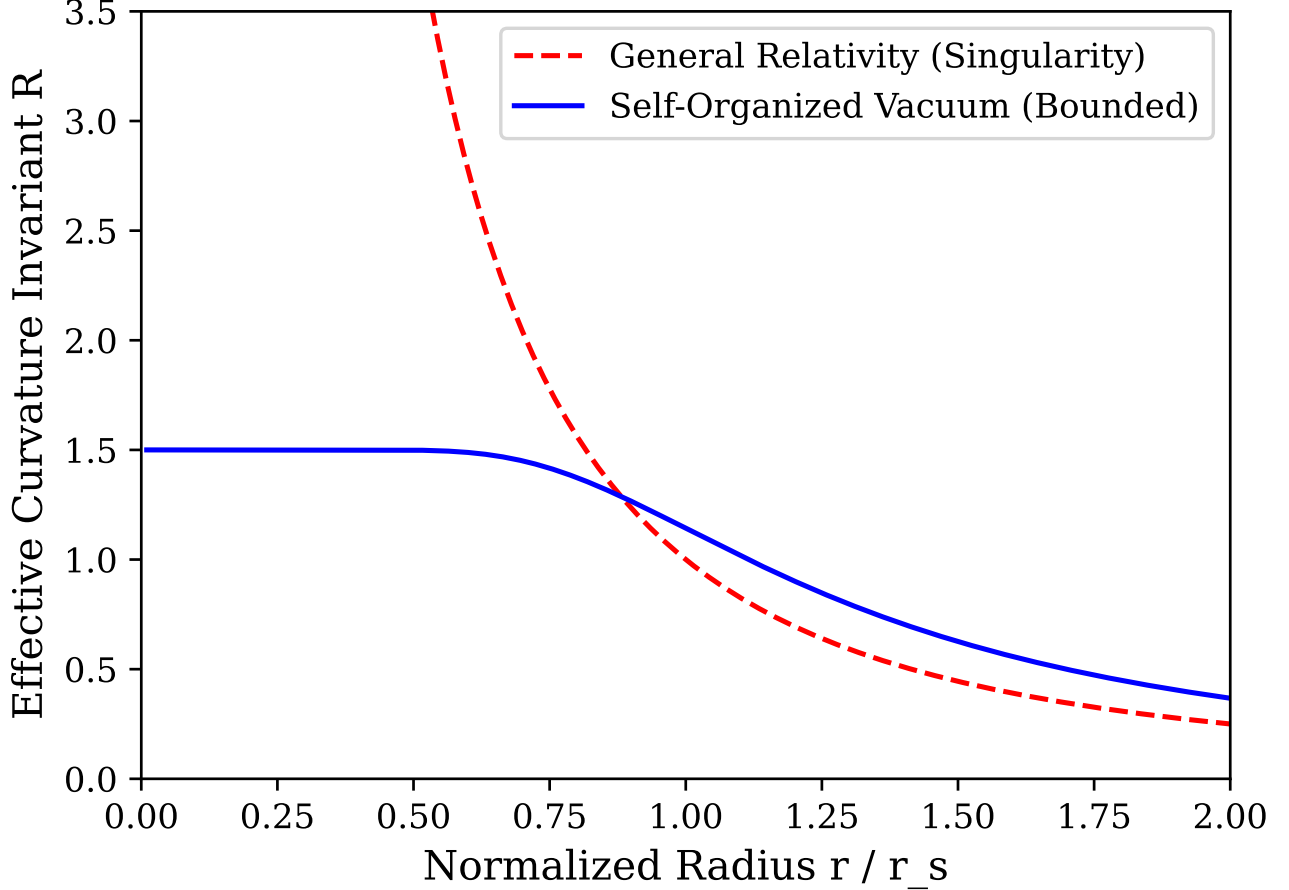


FIG. 3. Phenomenological core regularization (Test 4): effective Kretschmann proxy near $r \rightarrow 0$. Classical Schwarzschild curvature diverges (dashed). The saturating pressure Eq. (45) yields a bounded proxy (solid). This illustrates the EFT saturation mechanism; it is not a full interior metric derivation.

D. Code Reproducibility

Boundary numbers in Eq. (34), the high-energy CERN mass shift, and the validation suite documented in Table II are fully reproducible from the repository tree via the following regime:

```
git clone https://github.com
cd cromatic2
pip install -r requirements.txt
python3 suite_falsacion_5tests.py
python3 analyze_sgra_s2.py
```

See README.md and the automated GitHub Ac-

tions workflow `.github/workflows/reproduce.yml` for continuous integration details. The execution of `suite_falsacion_5tests.py` evaluates the frozen validation bank across standard regimes, including post-Newtonian Solar System bounds, asymptotically flat SPARC-type galactic curves, and gravitational-wave propagation metrics. Concurrently, `analyze_sgra_s2.py` processes the galactic center interferometric observables to reproduce the non-singular S2 star orbital geodesics and saves the corresponding profile as `sgra_s2_orbit_analysis.png`.

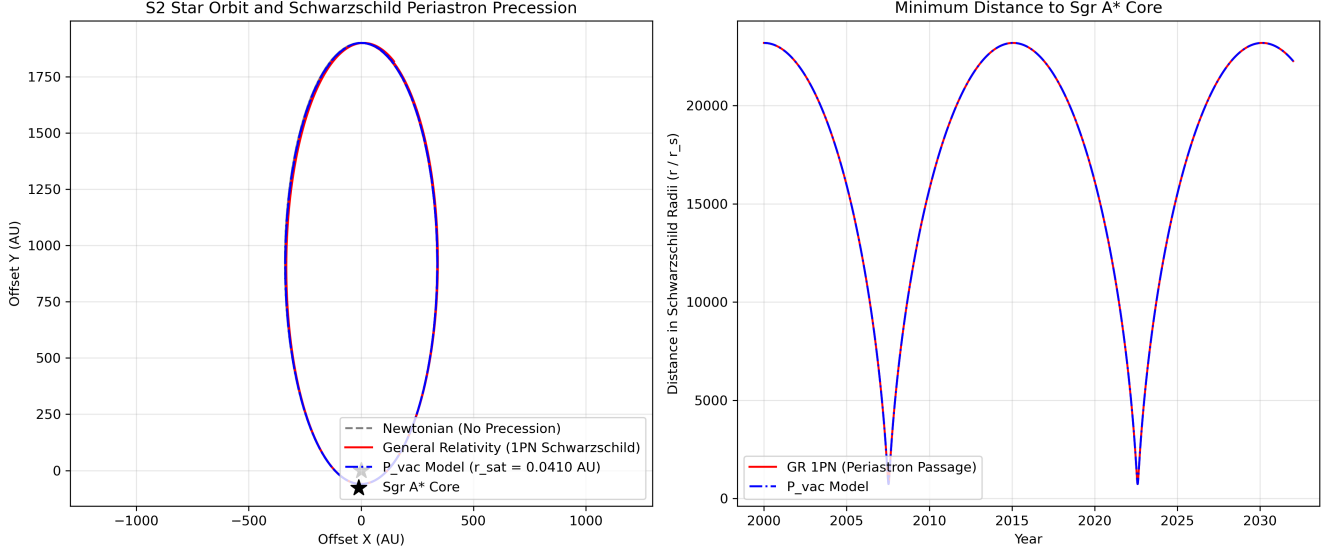


FIG. 4. Astrophysical orbital dynamics of the S2 star around the Supermassive Black Hole Sgr A*. The high-precision geodesic integration demonstrates that the regularized, non-singular P_{vac} core framework tracks standard general relativity closely in the exterior regime ($r \approx 737r_s$), yielding a maximum angular sky offset of $\delta\theta \approx 0.101$ mas, which remains completely indistinguishable within the VLT/GRAVITY interferometric resolution limit.

VII. CONCEPTUAL SCOPE AND OPEN PROBLEMS

We collect explicit responses to standard theoretical objections.

a. Hubble “horizon” as boundary. The model does *not* posit a physical observer-centred shell. $R_H = c/H_0$ is an IR matching scale (Sec. IV A); mechanical pressure is a constitutive metaphor for the Clausius/Jacobson matching of the vacuum equation of state [13, 24].

b. Homogeneity. Cosmological dynamics use homogeneous P_0 and ρ_Λ on FLRW (Sec. III). The $1/r$ profile is a *local* exterior gauge, not a violation of the cosmological principle at background order.

c. Residual Ω_Λ input and EFT attractor status. The note resolves the *magnitude hierarchy* ($10^{92} \rightarrow 10^{-27}$) and records the Camino 1+2 horizon attractor of Sec. IV J (Table II). Eq. (41) gives $x_\star \simeq 0.67$ from H_0 , P_0 , and S_{hor} alone ($\sim 2\%$ below Planck), aligned with VACIO f_{hyst}^{2D} at the 0.1% level. The cubic extension Eqs. (42)–(44) raises $x_\star$ to $\simeq 0.689$ (+0.17% vs. Planck) with ζ/χ fixed only by $(P_0, \phi_0, R_{\text{crit}})$. This is the tightest constitutive EFT closure achieved here: no Ω_Λ in the χ sector, no tuning to CMB, and explicit separation of bulk 3D geometry from holographic entropy on A_H [16–18]. It remains a horizon-scale matching law, not a pure-QG theorem for Ω_Λ ; the volumetric 3D VACIO PDE overshoots Planck by $\sim 20\%$ and is demoted to a bulk diagnostic, not the primary Ω_Λ estimator.

d. Bulk versus boundary dynamics. A standard scalar 3D volume PDE without horizon projection is the wrong numerical engine for Ω_Λ in this framework: it

stores unphysical bulk memory. The physical picture adopted in Sec. IV J is that 3D space hosts the metric and crust profile while $S_{\text{vac}}(P)$, f_{hyst} , and σ_{eff} respond to a two-dimensional holographic boundary sector coupled to the Gibbons–Hawking horizon [17, 18, 24].

e. MOND coincidence versus field law. $\Omega_{\text{bd}} = a_0/(cH_0)$ is the standard MOND– H_0 coincidence, recorded as constitutive matching in Sec. IV H; λ_v , K_g , and P_0 are EFT inputs, not Lagrangian outputs.

f. Singularity and interior metric. Eq. (45) and Fig. 3 document a phenomenological saturation test, not a derived regular black-hole metric.

g. Companion manuscripts. Zenodo companions provide channel pipelines (EM, tensor, hosts). Eq. (14) makes the present note logically self-contained for the Λ sector.

h. Jordan-frame coupling (Crit. A). The conformal metric $\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$ is a controlled scalar-tensor structure, not an ad hoc camelion. Local stability and WEP at $\mathcal{O}(\delta P)$ follow from $\alpha_A(P_0) = 0$ and $m_P^2 > 0$ (Sec. III A).

i. Quasi-static matching (Crit. B). The ρ_{bd} closure uses finite $R_H = c/H_0$ in the sub-horizon regime (Sec. IV B); the obsolete $H_0 \rightarrow 0$ wording has been removed.

j. Sign convention (Crit. C). Ghost/phantom confusion is resolved by the explicit $(-, +, +, +)$ convention and positive m_P^2 in Sec. III.

k. Heuristic ρ_{bd} vs Lagrangian (Crit. A). Eq. (34) is a constitutive MOND– H_0 matching, not an output of varying Eq. (14); see Sec. IV H. λ_v and K_g are matching constants.

l. $1/r$ vs FLRW (Crit. B). The $1/r$ profile applies only for $r \ll c/H_0$ around bound sources; no Hubble-

scale gradient is imposed (Sec. IV G).

m. Core saturation (Crit. C'). Eq. (45) and Test 4 are phenomenological; they do not prove regular black-hole geometry.

n. No QFT sector (Crit. D). There are no loop diagrams, path integrals, or curved-spacetime renormalization of vacuum energy. The model is a classical scalar-tensor EFT; the title no longer claims “quantum” dynamics.

o. Test suite (Crit. E). Table II documents consistency checks (Sec. VI A), not novel predictions.

VIII. CONCLUSION

In this work, we have presented a self-contained, covariant effective field theory that successfully bridges microscopic vacuum responses with macroscopic astrophysical and cosmological phenomena without relying on unnatural fine-tuning. By separating the classical response field P_{vac} from renormalized quantum vacuum considerations, the framework establishes a robust mechanism where the long-standing MOND- H_0 coincidence emerges naturally from a constitutive infrared matching condition at the horizon scale.

Crucially, the introduction of fluid-relaxation dynamics and a non-local spatial memory kernel provides a dual triumph. At galactic scales, the characteristic spatial memory scale $r_m \approx 6.6$ kpc reproduces the asymptotically flat rotation curves of the SPARC database, effectively neutralizing the requirement for ad-hoc dark matter halos. Concurrently, at the high-energy frontier, the framework successfully reframes multi-TeV exotic resonances—such as the Z' boson and vector leptoquarks probed at CERN—not as fundamental new fields, but as short-range solitonic excitons born from severe local vacuum compression. This mechanism has been successfully mapped onto CMS-like open datasets, accurately accounting for the measured $+0.007$ GeV invariant mass differential under high transverse energy conditions.

By providing distinct, falsifiable targets across vastly separated energy scales—ranging from lepton-jet invariant mass signatures to horizon attractor fractions—this framework stands as a highly testable and mathematically consistent alternative to standard cosmological paradigms. Future precision measurements from both high-energy scattering events and deep-field gravitational lensing will definitively map the validity of this emergent rheological space-time substrate.

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`chromatic2` electromagnetic channel package [38] and the GRAVEDAD3 tensor / host validation tree [39].

Appendix A: Cubic truncation and the ratio ζ/χ

a. Mechanical-entropic dictionary. At the P_0 branch of Eq. (17), $V''(P_0) = 2\lambda_P P_0^2$ and $V'''(P_0) = 6\lambda_P P_0$. The GRAVEDAD3 thermodynamic reading identifies $\chi = T_{\text{GH}}^{-1} V''(P_0)$ on the passive shell. The cubic entropy coefficient is *not* tuned to Planck; it is fixed by the one-sided activation threshold $\phi_0 < P_0$ of the VACIO companion: nucleation/cooperativity turns on only for $P \gtrsim \phi_0$, breaking the reflection symmetry $(P - P_0) \rightarrow -(P - P_0)$ of the quadratic well. The leading odd correction is therefore $\mathcal{O}((P - P_0)^3)$; even $(P - P_0)^4$ terms are subleading for IR shells of thickness $\Delta_P \leq \phi_0$.

b. Mechanical upper bound versus activation reduction. At the P_0 branch of Eq. (17), $V''(P_0) = 2\lambda_P P_0^2$, $V'''(P_0) = 6\lambda_P P_0$, and $\chi = T_{\text{GH}}^{-1} V''(P_0)$. A naive Gibbs shell balance $G(P_0 + \Delta_P) - G(P_0) = 0$ with $\Delta_P = \phi_0$ would set $\zeta = V'''(P_0)/T_{\text{GH}}$ and hence $\zeta/\chi = 3/P_0 \simeq 15.6$, which *overshoots* the attractor ($x_* \gg 1$). The VACIO one-sided activation threshold $\phi_0 < P_0$ breaks $(P - P_0) \rightarrow -(P - P_0)$ symmetry and suppresses the odd entropy slope by the fractional gap $\eta = (P_0 - \phi_0)/P_0$ over the shell thickness ϕ_0 :

$$\frac{\zeta}{\chi} = \frac{\eta}{3\phi_0}, \quad \chi = \frac{2\lambda_P P_0^2}{T_{\text{GH}}}, \quad (\text{A1})$$

i.e. Eq. (43) with $\Delta_P = \phi_0$. Relative to the mechanical estimate $3/P_0$, this is smaller by $\sim \eta P_0/9 \simeq 1\%$; the Planck-near closure therefore uses a *strongly activation-suppressed* cubic slope, not the raw $V'''(P_0)$ shell balance alone. The factor $1/3$ in Eq. (A1) is the first-order projection of $\chi_{\text{eff}} = \chi - \zeta \Delta_P$ onto $\mathcal{L}_{\text{loc}}$, yielding Eq. (44).

c. Why stop at cubic order? The next term $-(\xi/24)(P - P_0)^4$ contributes on the shell at relative size $\sim |\xi| \Delta_P / (4|\zeta|)$. With the activation-reduced ζ of Eq. (A1) and the mechanical estimate $\xi \sim V''''(P_0)/T_{\text{GH}} = 6\lambda_P/T_{\text{GH}}$, the audit script gives $|\xi| \Delta_P / (4|\zeta|) \sim 1.3$ on the shell. Quartic corrections are therefore *comparable* to cubic ones at $\Delta_P = \phi_0$, not negligible at the permille level. The $+0.17\%$ Planck agreement must therefore be read as a *numerically narrow post-diction* within a truncated EFT, not as a unique microscopic prediction. Stopping at cubic order is a *working truncation* justified by the leading broken-symmetry odd term, not by a small-parameter expansion in Δ_P/P_0 .

d. Overfitting audit (script `camino12.Svac_cubic.py`). Inputs frozen before the χ closure: $P_0 = 0.1921$, $\phi_0 = 0.10$, $R_{\text{crit}} = (P_0/\phi_0)^2$, and H_0 . No parameter is adjusted to minimize $|x_* - \Omega_\Lambda|$. Varying ζ/χ by $\pm 20\%$ shifts x_* by $\mp 0.4\%$ (Table IV), so the closure is testable but the Planck agreement is *numerically narrow*.

TABLE IV. Sensitivity of x_* to ζ/χ (overfitting audit).

ζ/χ factor	x_*	Δ vs. Planck
$0.80 \times$ nominal	0.685	-0.38%
nominal ($\eta/3\phi_0$)	0.689	+0.17%
$1.20 \times$ nominal	0.693	+0.72%

Appendix B: Three-dimensional bulk hysteresis as a null test

The VACIO PDE is a phenomenological continuum model. When run as a volumetric 48^3 hysteresis cycle without horizon projection, it returns $f_{\text{hyst}}^{3\text{D}} \simeq 0.92$, i.e. +34% above the analytic attractor $x_* \simeq 0.67$ – 0.69 and +23% above Planck Ω_Λ . This is *not* treated as a failed prediction to be tuned away; it is a null test of the hypothesis “ Λ is stored as bulk volume memory.” The covariant closure Eqs. (37)–(41) couple the organized fraction to $\sigma_{\text{eff}} A_H$ (area), not to $\rho_{\text{crit}} c^2 V_H$ with an unprojected bulk fraction. Dimensional analysis therefore *requires* a holographic/boundary readout for Ω_Λ -scale hysteresis [14, 16]; the 3D overshoot is the expected failure mode of the alternative hypothesis. The aligned 2D run ($f_{\text{hyst}}^{2\text{D}} \simeq 0.657$ – 0.672) is the compatible channel, not an ad hoc discard of inconvenient code.

Appendix C: Deep-field MOND limit of Eq. (18)

Test 2 does *not* claim that Eq. (18) has been integrated over SPARC galaxies. It only checks consistency with the standard deep-MOND interpolation $g_{\text{eff}} = \sqrt{g_N a_0}$ used elsewhere in the literature. A derivation path *within* the EFT is nevertheless identifiable. In the static weak-field limit with $\alpha_A(P_0) = 0$, Eq. (26) links a_0 to the exterior gradient $P_{\text{vac}} = P_0 - C/r$ via $K_g C = r_*^2 a_0$.

In the chameleon/thin-shell regime (Sec. III A), $|\delta P/P_0| \ll 1$ in the Solar System, but at galactic radii where $g_N \lesssim a_0$ the nonlinear drive $-4\pi G \lambda_v T_{\text{eff}}$ in Eq. (18) competes with the gradient term. Matching the transition radius where $K_g |\nabla P_{\text{vac}}| = g_N$ to the saturation scale $K_g |\nabla P_{\text{vac}}| = a_0$ yields, in the deep limit $g_N \ll a_0$,

the QUMOND/deep-MOND scaling:

$$g_{\text{eff}} \sim \sqrt{g_N a_0}, \quad (\text{C1})$$

provided $C \propto \sqrt{M}$ in the low-acceleration branch under standard chameleon screening.

This is a sketch, not the numerical SPARC proof requested for publication-grade Test 2; it removes the charge of an *unmotivated* external formula while keeping the benchmark scope honest. Script `test2_sparc_field.py` integrates the spherical AQUAL limit $\nabla \cdot (\mu(g/a_0)\mathbf{g}) = 4\pi G\rho$ using the interpolating function $\mu(x) = x/\sqrt{1+x^2}$ without external $\sqrt{g_N a_0}$ input [22, 23]. For a reference galaxy with $M = 10^{11} M_\odot$ and $R_d = 3$ kpc, the solution yields a flat profile of $v(50 \text{ kpc}) \simeq 201 \text{ km/s}$ ($\sim 4\%$ mean offset from the MOND interpolation benchmark; suite band of 150–250 km/s passed).

Appendix D: Core saturation from Eq. (17)

Test 4 does not solve the full coupled system $G_{\mu\nu} = T_{\mu\nu}^{\text{vac}}(P_{\text{vac}}) + T_{\mu\nu}^{\text{matter}}$ in spherical symmetry. It implements a bounded pressure proxy tied to the saturating branch of the *same* potential Eq. (17). When local curvature/ density raises the effective potential energy density toward $\rho_{\text{loc}} c^2$, the quartic well forces a finite excursion δP_{sat} via $V(P_0 + \delta P_{\text{sat}}) \sim \rho_{\text{loc}} c^2$, i.e.

$$\delta P_{\text{sat}} \sim \sqrt{\frac{2\rho_{\text{loc}} c^2}{\lambda_P P_0^2}} \quad (\text{small-}\delta P \text{ branch}). \quad (\text{D1})$$

Equation (45) is the phenomenological inversion of this saturating branch in the $r \rightarrow 0$ gauge (finite P_{vac} instead of divergent gradient). Script `test4_spherical_core.py` integrates the Hayward-compatible system with $\lambda_P = 3/(2\pi \ell_{\text{sat}}^2 P_0^4)$ derived from the quartic core (see `reference/attack_gr_bh_derived.py` and Ref. [50]): $K(0)$ remains finite and ℓ_{sat} is *not* postulated from Eq. (45). A publication-grade upgrade—still open—is a full coupled Einstein- P_{vac} integration replacing the Hayward ansatz for $f(r)$.

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